

Single-Qubit Gates

Rotate a state anywhere on the sphere. Pauli X, Y, Z, the phase gates S and T, and the RX/RY/RZ family as continuous rotations.

EXAMPLE NOTES

Written by the QuantaVerse team to show what a module's notes can look like: each lesson's theory in brief, every formula typeset and numbered, and the key ones gathered on a formula sheet at the end. Professors can upload their own notes for this module on its page, and students will see them alongside these.

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CONVENTIONS

Matrices act on column vectors in the basis ($|0\rangle$, $|1\rangle$). A circuit read left to right applies its gates in that order, so its matrix is written right to left. $U \equiv V$ means equal up to a global phase.

LESSON 3.1

The Pauli gates

KEY IDEA

X , Y and Z are half-turns of the Bloch sphere about its three axes. Each squares to the identity, and any two of them anticommute.

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (3.1.1)$$

the Pauli matrices

X is the quantum NOT: it swaps $|0\rangle$ and $|1\rangle$. Z leaves $|0\rangle$ alone and flips the sign of $|1\rangle$, a *phase flip* that is invisible to a Z measurement. Y does both, with a factor of i .

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle, \quad Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle, \quad Y|0\rangle = i|1\rangle, \quad Y|1\rangle = -i|0\rangle \quad (3.1.2)$$

bit flip, phase flip, and both

$$X^2 = Y^2 = Z^2 = I, \quad XY = iZ, \quad YZ = iX, \quad ZX = iY \quad (3.1.3)$$

the Pauli algebra

$$\{\sigma_j, \sigma_k\} = \sigma_j \sigma_k + \sigma_k \sigma_j = 2\delta_{jk} I \quad (3.1.4)$$

distinct Paulis anticommute: $XZ = -ZX$

Each Pauli has eigenvalues $+1$ and -1 , and its eigenvectors are the two poles of its own axis. That is why they double as *observables*: measuring Z reports ± 1 , and its average is the z coordinate of the Bloch vector.

GATE	EIGENVALUE $+1$	EIGENVALUE -1	ROTATION
X	$ +\rangle$	$ -\rangle$	180° about x
Y	$ +i\rangle$	$ -i\rangle$	180° about y
Z	$ 0\rangle$	$ 1\rangle$	180° about z

WORKED EXAMPLE · CHECKING ONE PRODUCT

$$XY = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = i \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = iZ.$$

In the other order, $YX = -iZ$. Up to a phase, XY and YX are both Z : two half-turns about perpendicular axes make a half-turn about the third.

LESSON 3.2

Why gates must be unitary

KEY IDEA

A gate must keep every state normalised, for every input. The matrices that do that are exactly the unitary ones, and every one of them can be undone.

$$U^\dagger U = UU^\dagger = I \quad (3.2.1)$$

unitarity

If $U^\dagger U = I$, then U preserves inner products, so lengths and angles between states are unchanged. In particular the probabilities of any measurement still add to one after the gate.

$$\langle U\phi | U\psi \rangle = \langle \phi | U^\dagger U | \psi \rangle = \langle \phi | \psi \rangle \quad (3.2.2)$$

unitaries preserve inner products

Unitarity also means reversibility: the inverse of a gate is its conjugate transpose, which is itself a valid gate. A circuit is undone by running the inverses of its gates in reverse order.

$$(U_k \cdots U_2 U_1)^\dagger = U_1^\dagger U_2^\dagger \cdots U_k^\dagger \quad (3.2.3)$$

undoing a circuit

$$U|v\rangle = \lambda|v\rangle \Rightarrow |\lambda| = 1, \lambda = e^{i\theta} \quad (3.2.4)$$

the eigenvalues of a unitary are pure phases

DEFINITION · HERMITIAN AND UNITARY

A Hermitian matrix satisfies $A^\dagger = A$ and represents something measurable. A unitary satisfies $U^\dagger = U^{-1}$ and represents something you can do. The Paulis and H are both at once, which is why each is its own inverse.

WORKED EXAMPLE · UNITARY OR NOT?

H : $H^\dagger H = H^2 = I$, so yes.

$A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ sends both $|0\rangle$ and $|1\rangle$ to $|0\rangle$. Two different inputs, one output: it cannot be undone and cannot be a gate. $A^\dagger A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \neq I$ confirms it.

Measurement and reset are not unitary, and that is why circuits treat them as separate operations rather than gates.

LESSON 3.3

Gates as rotations

KEY IDEA

Up to a global phase, every single-qubit unitary is a rotation of the Bloch sphere by some angle about some axis.

$$R_{\hat{n}}(\theta) = e^{-i\theta \hat{n} \cdot \vec{\sigma}/2} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} (n_x X + n_y Y + n_z Z) \quad (3.3.1)$$

rotation by θ about the unit axis $\hat{n}$

$$U = e^{i\alpha} R_{\hat{n}}(\theta) \quad (3.3.2)$$

every single-qubit gate, for some phase α , axis $\hat{n}$ and angle θ

The fixed gates of the palette are all half-turns, $\theta = \pi$, about different axes. With $\theta = \pi$ the formula gives $R_{\hat{n}}(\pi) = -i(\hat{n} \cdot \vec{\sigma})$, so each gate is its rotation times the global phase i .

GATE	AXIS $\hat{n}$	ANGLE	EXACTLY
X	$(1,0,0)$	π	$X = iR_x(\pi)$
Y	$(0,1,0)$	π	$Y = iR_y(\pi)$
Z	$(0,0,1)$	π	$Z = iR_z(\pi)$
H	$\frac{1}{\sqrt{2}}(1,0,1)$	π	$H = iR_{\hat{n}}(\pi)$

$$R_{\hat{n}}(\pi) = -i \frac{X+Z}{\sqrt{2}} = -iH \quad \text{for } \hat{n} = \frac{1}{\sqrt{2}}(1,0,1) \quad (3.3.3)$$

the Hadamard is a half-turn about the diagonal

Any rotation can be built from rotations about just two fixed axes. The standard choice is z, y, z , and this is the form compilers use to turn an arbitrary gate into hardware pulses.

$$U = e^{i\alpha} R_z(\beta) R_y(\gamma) R_z(\delta) \quad (3.3.4)$$

the Z-Y-Z decomposition

WORKED EXAMPLE · WHERE DOES X SEND A STATE?

X is a half-turn about x , so it maps the Bloch vector (r_x, r_y, r_z) to $(r_x, -r_y, -r_z)$. For $|+i\rangle$, at $(0,1,0)$, the result is $(0, -1, 0) = |-i\rangle$. Directly: $X \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle) = \frac{1}{\sqrt{2}}(|1\rangle + i|0\rangle) = \frac{i}{\sqrt{2}}(|0\rangle - i|1\rangle) \equiv |-i\rangle$.

LESSON 3.4

S and T — the phase gates

KEY IDEA

S and T add a relative phase of 90° and 45° to $|1\rangle$. They are quarter and eighth turns about z.

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} \quad (3.4.1)$$

the phase gates

$$T^2 = S, \quad S^2 = Z, \quad Z^2 = I, \quad T^8 = I \quad (3.4.2)$$

each is the square root of the one before

$$P(\varphi) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{pmatrix} = e^{i\varphi/2} R_z(\varphi) \quad (3.4.3)$$

the general phase gate: $S = P(\pi/2)$, $T = P(\pi/4)$, $Z = P(\pi)$

On a state on the equator, a phase gate is a turn about z by its angle. *S* takes $|+\rangle$ to $|+i\rangle$, a quarter turn; *T* takes it halfway there.

$$S|+\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}} = |+i\rangle, \quad T|+\rangle = \frac{|0\rangle + e^{i\pi/4}|1\rangle}{\sqrt{2}} \quad (3.4.4)$$

quarter and eighth turns of $|+\rangle$

$$S^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}, \quad T^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & e^{-i\pi/4} \end{pmatrix} \quad (3.4.5)$$

the inverses turn the other way

The difference between the two shows up when conjugating a Pauli. *S* turns *X* into *Y*, another Pauli. *T* turns *X* into a mixture of *X* and *Y*, which is not a Pauli. That property, keeping Paulis as Paulis or not, is the dividing line of lesson 3.7.

$$SXS^\dagger = Y, \quad TXT^\dagger = \frac{X+Y}{\sqrt{2}} \quad (3.4.6)$$

S maps Paulis to Paulis; *T* does not

WORKED EXAMPLE · VERIFYING $SXS^\dagger = Y$

$$SXS^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = Y.$$

LESSON 3.5

Continuous rotations

KEY IDEA

R_x , R_y and R_z take an angle, so one gate can reach any amount of rotation about its axis. The fixed gates are special angles of them.

$$R_x(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ -i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}, \quad R_y(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix} \quad (3.5.1)$$

rotations about x and y

$$R_z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix} \quad (3.5.2)$$

rotation about z

$$R_z(\pi) = -iZ \equiv Z, \quad R_z\left(\frac{\pi}{2}\right) = e^{-i\pi/4}S \equiv S, \quad R_z\left(\frac{\pi}{4}\right) = e^{-i\pi/8}T \equiv T \quad (3.5.3)$$

the fixed phase gates, as angles of R_z

Rotations about the same axis add their angles. A full turn, $\theta = 2\pi$, gives $-I$: the same physical operation as doing nothing, but a sign that matters once the rotation is controlled. Only $\theta = 4\pi$ gives exactly I .

$$R_{\hat{n}}(\theta_1) R_{\hat{n}}(\theta_2) = R_{\hat{n}}(\theta_1 + \theta_2), \quad R_{\hat{n}}(2\pi) = -I \quad (3.5.4)$$

angles add; a full turn is a sign

R_y is the useful one for preparing states with real amplitudes: from $|0\rangle$ it produces any probability split without adding a phase.

$$R_y(\theta)|0\rangle = \cos \frac{\theta}{2}|0\rangle + \sin \frac{\theta}{2}|1\rangle \Rightarrow P(1) = \sin^2 \frac{\theta}{2} \quad (3.5.5)$$

preparing a chosen probability

WORKED EXAMPLE · A 90/10 SPLIT

To get $P(1) = 0.1$, solve $\sin^2 \frac{\theta}{2} = 0.1$: $\theta = 2 \arcsin \sqrt{0.1} \approx 2 \times 0.3218 = 0.6435$ rad, about 36.9° .

Then $R_y(0.6435)|0\rangle \approx 0.949|0\rangle + 0.316|1\rangle$, and $0.949^2 \approx 0.9$.

Parametrised rotations are what variational algorithms tune: the angles are the parameters a classical optimiser adjusts between runs.

LESSON 3.6

Order matters

KEY IDEA

A circuit is read left to right, but its matrix is written right to left, and swapping two gates usually changes the result.

$$|\psi\rangle \rightarrow A \rightarrow B \iff BA|\psi\rangle \quad (3.6.1)$$

the first gate in time is the rightmost matrix

Matrix multiplication is not commutative, and neither are gates. The *commutator* measures how far two gates are from commuting; when it vanishes, their order does not matter.

$$[A, B] = AB - BA \quad (3.6.2)$$

the commutator: zero exactly when order does not matter

Conjugating by H swaps the roles of x and z , which explains most of the identities used in circuit simplification.

$$HXH = Z, \quad HZH = X, \quad HYH = -Y \quad (3.6.3)$$

conjugating by H swaps x and z , and reverses y

THE SAME TWO GATES, IN BOTH ORDERS

$$\begin{aligned} |0\rangle \xrightarrow{H} \xrightarrow{Z} &= Z \cdot H|0\rangle = Z|+\rangle = |-\rangle \\ |0\rangle \xrightarrow{Z} \xrightarrow{H} &= H \cdot Z|0\rangle = H|0\rangle = |+\rangle \end{aligned}$$

The two outputs above are genuinely different states, told apart with certainty by an X-basis measurement. Some reorderings do only change a global phase, which no measurement sees. Distinct Paulis are the standard case, since they anticommute.

$$XZ = -ZX \Rightarrow XZ|\psi\rangle \equiv ZX|\psi\rangle \quad (3.6.4)$$

anticommuting gates differ only by a sign

DEFINITION · COMMUTING GATES

Gates on different qubits always commute, and so do gates that are diagonal in the same basis, such as Z, S, T and R_z . A compiler may reorder those freely and merge runs of them into one rotation.

WORKED EXAMPLE · MERGING A RUN

$TSZ = P(\frac{\pi}{4})P(\frac{\pi}{2})P(\pi) = P(\frac{7\pi}{4}) = T^\dagger$, because phase gates about the same axis add their angles modulo 2π : three gates become one.

LESSON 3.7

T-depth, and what a gate costs

KEY IDEA

Clifford gates (H , S , $CNOT$ and the Paulis) are cheap to run fault-tolerantly and easy to simulate. T is neither, and the number of T gates is what a circuit really costs.

DEFINITION · CLIFFORD GROUP

The gates generated by H , S and $CNOT$. They map every Pauli operator to another Pauli operator, up to a sign, under conjugation.

$$CPC^\dagger \in \pm\{I, X, Y, Z\}^{\otimes n} \quad \text{for every Pauli string } P \quad (3.7.1)$$

what makes a gate Clifford

Because Cliffords only shuffle Paulis, a classical computer can track a Clifford circuit by tracking a few Pauli strings instead of 2^n amplitudes. Any circuit made only of them can therefore be simulated efficiently. This is the *Gottesman–Knill theorem*.

Adding T breaks that shortcut, and Clifford + T is universal: any single-qubit gate can be approximated to accuracy ε with a short sequence of them.

$$N_{\text{gates}} = O\left(\log^c \frac{1}{\varepsilon}\right) \quad (3.7.2)$$

the Solovay–Kitaev theorem: c is a small constant, about 4 in the textbook construction and lower in later ones

For z -rotations, optimal Clifford + T synthesis does better still, using about $3 \log_2(1/\varepsilon)$ T gates.

On an error-corrected machine the price difference is large. Clifford gates can be applied almost directly to encoded qubits, while each T needs a *magic state*, prepared and purified by a separate, costly distillation process. Circuits are therefore compared by two numbers.

MEASURE	COUNTS	WHY IT MATTERS
T-count	T and $T^\dagger$ gates in total	one magic state each
T-depth	layers containing a T	how long the magic-state supply must keep up
Clifford count	everything else	cheap by comparison

WORKED EXAMPLE · A TOFFOLI'S PRICE

The Toffoli gate (controlled-controlled-NOT) needs seven T or $T^\dagger$ gates in its standard decomposition into Clifford + T . A circuit with a thousand Toffolis therefore needs about seven thousand magic states before any other cost is counted.

Resource estimates for large algorithms, such as the Shor estimates in lesson 8.8, are dominated by T-count for exactly this reason.

MODULE 3 · FORMULA SHEET

Formula sheet

The key results of the module, numbered as they appear in the lessons.

the Pauli matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (3.1.1)$$

the Pauli algebra

$$X^2 = Y^2 = Z^2 = I, \quad XY = iZ, \quad YZ = iX, \quad ZX = iY \quad (3.1.3)$$

unitarity

$$U^\dagger U = U U^\dagger = I \quad (3.2.1)$$

unitaries preserve inner products

$$\langle U\phi | U\psi \rangle = \langle \phi | U^\dagger U | \psi \rangle = \langle \phi | \psi \rangle \quad (3.2.2)$$

rotation by θ about the unit axis $\hat{n}$

$$R_{\hat{n}}(\theta) = e^{-i\theta \hat{n} \cdot \vec{\sigma}/2} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} (n_x X + n_y Y + n_z Z) \quad (3.3.1)$$

every single-qubit gate, for some phase α , axis $\hat{n}$ and angle θ

$$U = e^{i\alpha} R_{\hat{n}}(\theta) \quad (3.3.2)$$

the Z-Y-Z decomposition

$$U = e^{i\alpha} R_z(\beta) R_y(\gamma) R_z(\delta) \quad (3.3.4)$$

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S maps Paulis to Paulis; T does not

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rotations about x and y

$$R_x(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ -i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}, \quad R_y(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix} \quad (3.5.1)$$

rotation about z

$$R_z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix} \quad (3.5.2)$$

angles add; a full turn is a sign

$$R_{\hat{n}}(\theta_1) R_{\hat{n}}(\theta_2) = R_{\hat{n}}(\theta_1 + \theta_2), \quad R_{\hat{n}}(2\pi) = -I \quad (3.5.4)$$

the first gate in time is the rightmost matrix

$$|\psi\rangle \rightarrow A \rightarrow B \quad \Leftrightarrow \quad BA|\psi\rangle \quad (3.6.1)$$

conjugating by H swaps x and z , and reverses y

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$$CPC^\dagger \in \pm\{I, X, Y, Z\}^{\otimes n} \quad \text{for every Pauli string } P \quad (3.7.1)$$

the Solovay-Kitaev theorem: c is a small constant, about 4 in the textbook construction and lower in later ones

$$N_{\text{gates}} = O\left(\log^c \frac{1}{\varepsilon}\right) \quad (3.7.2)$$